Printing Inks

Product / Application	BIS	ISO	ASTM	DIN	Other Standards(SSPC,AMPP, REACH, FDA)
Offset Printing Ink	IS 15495; IS 6931	ISO 2846-1 (Colour and Transparency of Printing Ink); ISO 12647 (Process Control for Printing),ISO 2834	ASTM D2066 (Printing Ink Evaluation); ASTM D523 (Gloss Measurement); ASTM D2244 (Colour Difference); ASTM D3359 (Adhesion); ASTM D5264 (Abrasion Resistance of Printed Materials); ASTM D3455 (Compatibility of Printing Inks with Substrates where applicable),ASTM D1310; ASTM D1475; ASTM D4040; ASTM D4302	DIN ISO 12647 series (Graphic Technology — Process Control for Offset Printing); DIN ISO 2846-1 (Colour and Transparency of Ink Sets); DIN EN ISO 2836 (Resistance of Prints to Chemicals); DIN EN ISO 11664 series (Colour Measurement); DIN EN ISO 2813 (Gloss of Coatings/Printed Surfaces)	REACH: Regulation (EC) No. 1907/2006 (inks solvents, pigments, heavy metals, SVHC substances); FDA: applicable only for food-contact packaging inks under FDA indirect food additive regulations
Flexographic Ink	IS 15495; IS 6931	ISO 2846-5 (Colour and Transparency of Flexographic Ink); ISO 12647 Series,ISO 2834	ASTM D2066 (Printing Ink Testing); ASTM D5264 (Abrasion Resistance of Printed Materials); ASTM D2244 (Colour Difference); ASTM D523 (Gloss); ASTM D3359 (Ink Adhesion),ASTM D1310; ASTM D1475; ASTM D4040; ASTM D4302	DIN ISO 12647-6 (Flexographic Process Control); DIN ISO 2846-5 (Colour and Transparency of Ink Sets for Flexographic Printing); DIN EN ISO 2836 (Chemical Resistance of Prints); DIN EN ISO 11664 (Colour Measurement); DIN EN ISO 105-A02 (Grey Scale Evaluation where textile-related applications apply)	REACH: Regulation (EC) No. 1907/2006 (solvents, pigments, additives); FDA: FDA 21 CFR 175.105 (Adhesives) / 21 CFR 176.170 & 176.180 where ink systems are used in food packaging structures
Gravure Printing Ink	IS 15495; IS 6931	ISO 2846 Series (Printing Ink Colour); ISO 12647 Series,ISO 2834	ASTM D2066 (Printing Ink Evaluation); ASTM D5264 (Abrasion Resistance); ASTM D2244 (Colour Measurement); ASTM D3359 (Adhesion); ASTM D523 (Gloss),ASTM D1310; ASTM D1475; ASTM D4040; ASTM D4302	DIN ISO 12647-4 (Gravure Process Control); DIN ISO 2846 series (Ink Colour and Transparency); DIN EN ISO 2836 (Resistance of Prints); DIN EN ISO 11664 (Colour Measurement); DIN EN ISO 2813 (Gloss Measurement)	REACH: Regulation (EC) No. 1907/2006 (solvents, pigments, VOC, restricted substances); FDA: applicable for food packaging printing applications
Screen Printing Ink	IS 15495; IS 6931	ISO 12647 Series (Graphic Technology Process Control); ISO 2846 Series,ISO 2834	ASTM D2066 (Printing Ink Testing); ASTM D3359 (Adhesion); ASTM D4060 (Abrasion Resistance); ASTM D523 (Gloss); ASTM D2244 (Colour Difference),ASTM D1310; ASTM D1475; ASTM D4040; ASTM D4302	DIN ISO 12647-5 (Screen Printing Process Control); DIN EN ISO 2836 (Chemical Resistance of Prints); DIN EN ISO 11664 (Colour Measurement); DIN EN ISO 2409 (Adhesion of Coating/Ink Films); DIN EN ISO 2813 (Gloss)	REACH: Regulation (EC) No. 1907/2006 (pigments, solvents, additives); FDA: only where printed surface is food-contact related
Digital Printing Ink	IS 15495; IS 6931	ISO 12647 Series (Graphic Technology Process Control); ISO 2846 Series where applicable,ISO 2834	ASTM D2066 (Printing Ink Testing); ASTM D2244 (Colour Difference); ASTM E308 (Spectrophotometric Colour Measurement); ASTM D3359 (Adhesion); ASTM D5264 (Abrasion Resistance),ASTM D1310; ASTM D1475; ASTM D4040; ASTM D4302	DIN ISO 12647 series (Digital Printing Process Control where applicable); DIN ISO 13655 (Spectral Measurement of Prints); DIN ISO 12640 series (Colour Data); DIN EN ISO 11664 (Colour Measurement); DIN EN ISO 2836 (Print Resistance)	REACH: Regulation (EC) No. 1907/2006 (inks, solvents, pigments, photoinitiators); FDA: applicable only for food packaging printing applications
Inkjet Ink	IS 15495; IS 6931	ISO 12647 Series (Printing Process Control); ISO 2846 Series where applicable,ISO 2834	ASTM D2066 (Printing Ink Evaluation); ASTM D2244 (Colour Difference); ASTM E308 (Colour Measurement); ASTM D3359 (Adhesion); ASTM D5264 (Abrasion Resistance),ASTM D1310; ASTM D1475; ASTM D4040; ASTM D4302	DIN ISO 13655 (Spectral Measurement of Prints); DIN ISO 12647 series (Digital Print Quality); DIN EN ISO 11664 (Colour Measurement); DIN EN ISO 2836 (Chemical Resistance); DIN EN ISO 105-X12 (Colour Fastness to Rubbing where textile inkjet applications apply)	REACH: Regulation (EC) No. 1907/2006 (solvents, dyes, pigments, additives); FDA: applicable only for food-contact printing applications
UV Curable Ink	IS 15495; IS 6931	ISO 12647 Series; ISO 2813 (Gloss Measurement); ISO 2409 (Adhesion),ISO 2834	ASTM D7027 (UV-Cured Coating Characterization where applicable); ASTM D5895 (Curing/Drying Time); ASTM D3359 (Adhesion); ASTM D523	DIN EN ISO 2836 (Resistance of Prints); DIN ISO 12647 (Print Process Control); DIN ISO 13655 (Spectral Measurement); DIN EN ISO 16474	REACH: Regulation (EC) No. 1907/2006 (acrylates, photoinitiators, reactive monomers); FDA: applicable for food packaging

Product / Application	BIS	ISO	ASTM	DIN	Other Standards(SSPC,AMPP, REACH, FDA)
			(Gloss); ASTM D4060 (Abrasion Resistance),ASTM D1310; ASTM D1475; ASTM D2066; ASTM D4040; ASTM D4302	(Artificial Weathering); DIN EN ISO 2813 (Gloss); DIN EN ISO 2409 (Adhesion)	inks where compliant formulation is used
Solvent Based Printing Ink	IS 15495; IS 6931	ISO 2846 Series; ISO 12647 Series,ISO 2834	ASTM D2066 (Printing Ink Testing); ASTM D2369 (Volatile Content); ASTM D2244 (Colour Difference); ASTM D3359 (Adhesion); ASTM D5264 (Abrasion Resistance),ASTM D1310; ASTM D1475; ASTM D4040; ASTM D4302	DIN ISO 12647 series; DIN ISO 2846 series; DIN EN ISO 2836; DIN EN ISO 11664; DIN EN ISO 11890-2 (VOC Content Determination where applicable)	REACH: Regulation (EC) No. 1907/2006 (VOC solvents, pigments, additives); FDA: only for food packaging applications
Water Based Printing Ink	IS 15495; IS 6931	ISO 2846 Series; ISO 12647 Series,ISO 2834	ASTM D2066 (Printing Ink Testing); ASTM D2369 (Volatile Content); ASTM D2244 (Colour Difference); ASTM D3359 (Adhesion); ASTM D5264 (Abrasion Resistance),ASTM D1310; ASTM D1475; ASTM D4040; ASTM D4302	DIN ISO 12647; DIN ISO 2846; DIN EN ISO 2836; DIN EN ISO 11664; DIN EN ISO 11890 (VOC Determination where applicable)	REACH: Regulation (EC) No. 1907/2006 (binders, pigments, preservatives); FDA: only for food packaging applications
LED Curable Ink	IS 15495; IS 6931	ISO 12647 Series; ISO 2813 (Gloss); ISO 2409 (Adhesion),ISO 2834	ASTM D5895 (Drying/Curing Time); ASTM D7027 (UV-Cured Coating Characterization); ASTM D3359 (Adhesion); ASTM D4060 (Abrasion Resistance),ASTM D1310; ASTM D1475; ASTM D2066; ASTM D4040; ASTM D4302	DIN ISO 13655 (Spectral Measurement); DIN ISO 12647 (Process Control); DIN EN ISO 2836 (Chemical Resistance); DIN EN ISO 2409 (Adhesion); DIN EN ISO 16474 (Weathering)	REACH: Regulation (EC) No. 1907/2006 (photoinitiators, acrylates, additives); FDA: applicable only for compliant food packaging printing inks
Packaging Ink	IS 15495; IS 6931	ISO 12647 Series; ISO 2846 Series; ISO 1186 Series (Migration Testing where food contact applies),ISO 2834	ASTM D2066 (Printing Ink Evaluation); ASTM D3359 (Adhesion); ASTM D5264 (Abrasion Resistance); ASTM D1308 (Chemical Resistance); ASTM F1249 (Water Vapour Transmission where barrier packaging applies),ASTM D1310; ASTM D1475; ASTM D4040; ASTM D4302	DIN EN 646 (Paper and Board Intended for Food Contact — Colour Fastness); DIN EN 1541 (Determination of Transfer of Substances from Paper and Board); DIN ISO 12647; DIN ISO 2846; DIN EN ISO 2836	FDA: 21 CFR 175.105 (adhesive-related packaging structures where applicable); FDA 21 CFR 176 series (paper and paperboard food-contact applications); REACH: Regulation (EC) No. 1907/2006
Food Packaging Ink	IS 15495; IS 6931	ISO 12647 (Graphic Technology Process Control); ISO 1186 Series (Migration Testing where applicable),ISO 2834	ASTM D2066 (Printing Ink Testing); ASTM D3359 (Adhesion); ASTM D5264 (Abrasion Resistance); ASTM D543 (Chemical Resistance of Plastics where applicable); ASTM F1249 (WVTR where applicable),ASTM D1310; ASTM D1475; ASTM D4040; ASTM D4302	DIN EN 646 (Colour Fastness of Paper and Board); DIN EN 14338 (Paper and Board for Food Contact); DIN EN 1186 series (Food Contact Materials); DIN EN 13130 series (Specific Migration); DIN ISO 12647	FDA: 21 CFR 175.105; FDA 21 CFR 176.170; FDA 21 CFR 176.180; REACH: Regulation (EC) No. 1907/2006 (SVHC, restricted substances, heavy metals)
Newspaper Ink	IS 15495; IS 6931	ISO 2846 Series; ISO 12647-3 (Coldset Offset Printing Process Control),ISO 2834	ASTM D2066 (Printing Ink Evaluation); ASTM D523 (Gloss); ASTM D2244 (Colour Difference); ASTM D5264 (Abrasion Resistance),ASTM D1310; ASTM D1475; ASTM D4040; ASTM D4302	DIN ISO 12647-3 (Coldset Offset Printing Process Control); DIN ISO 2846-1 (Ink Colour and Transparency); DIN EN ISO 2836 (Print Resistance); DIN ISO 13655 (Spectral Measurement)	REACH: Regulation (EC) No. 1907/2006 (mineral oils, pigments, solvents, additives)
Publication Ink	IS 15495; IS 6931	ISO 2846 Series; ISO 12647 Series,ISO 2834	ASTM D2066 (Printing Ink Testing); ASTM D2244 (Colour Measurement); ASTM D523 (Gloss); ASTM D5264 (Abrasion Resistance),ASTM D1310; ASTM D1475; ASTM D4040; ASTM D4302	DIN ISO 12647 series; DIN ISO 2846 series; DIN ISO 13655; DIN EN ISO 2836; DIN EN ISO 2813	REACH: Regulation (EC) No. 1907/2006 (pigments, solvents, VOC, additives)
Security Printing Ink	IS 15495; IS 6931	ISO 12647 Series; ISO 2846 Series,ISO 2834	ASTM D2066 (Printing Ink Evaluation); ASTM D2244 (Colour Difference); ASTM E308 (Colour Measurement); ASTM D523 (Gloss); Application-specific optical/security testing where applicable,ASTM D1310; ASTM D1475; ASTM D4040; ASTM D4302	DIN ISO 12647 (Print Process Control); DIN ISO 13655 (Spectral Measurement); DIN EN ISO 11664 (Colour Measurement); DIN EN ISO 12944 where security coatings apply; application-specific Bundesdruckerei/security requirements where applicable	REACH: Regulation (EC) No. 1907/2006 (pigments, magnetic materials, additives, restricted substances)

Product / Application	BIS	ISO	ASTM	DIN	Other Standards(SSPC,AMPP, REACH, FDA)
Barcode Ink	IS 15495; IS 6931	ISO 15416 (Barcode Print Quality); ISO 12647 Series,ISO 2834	ASTM D2066 (Ink Testing); ASTM D2244 (Colour Difference); ASTM E308 (Colour Measurement); ASTM D5264 (Abrasion Resistance); ASTM D3951 (Packaging Marking Applications where applicable),ASTM D1310; ASTM D1475; ASTM D4040; ASTM D4302	DIN ISO/IEC 15416 (Barcode Print Quality — Linear Symbols); DIN ISO/IEC 15415 (2D Barcode Print Quality); DIN ISO 13655 (Print Measurement); DIN ISO 12647	REACH: Regulation (EC) No. 1907/2006 (pigments, solvents, additives)
Textile Printing Ink	IS 15495; IS 6931	ISO 105 Series (Colour Fastness); ISO 12647 Series,ISO 2834	ASTM D2066 (Ink Testing); ASTM D5264 (Abrasion Resistance); ASTM D3359 (Adhesion where applicable); ASTM D4966 (Abrasion Resistance of Textile Materials); ASTM D5034 (Breaking Strength of Textile Fabrics where applicable),ASTM D1310; ASTM D1475; ASTM D4040; ASTM D4302	DIN EN ISO 105 series (Textile Colour Fastness); DIN EN ISO 105-X12 (Rubbing Fastness); DIN EN ISO 105-C06 (Wash Fastness); DIN EN ISO 105-E04 (Perspiration Fastness); DIN EN ISO 11664 (Colour Measurement)	REACH: Regulation (EC) No. 1907/2006; REACH Annex XVII (restricted azo substances); FDA: —
Ceramic Printing Ink	IS 15495; IS 6931	ISO 10545 Series (Ceramic Testing); ISO 12647 Series where applicable,ISO 2834	ASTM C242 (Ceramic Whitewares Terminology); ASTM C1026 (Freeze-Thaw Resistance where applicable); ASTM E308 (Colour Measurement); ASTM D3359 (Adhesion where coating application applies),ASTM D1310; ASTM D1475; ASTM D2066; ASTM D4040; ASTM D4302	DIN EN ISO 10545 series (Ceramic Tiles Testing where applicable); DIN EN ISO 11664 (Colour Measurement); DIN EN ISO 2813 (Gloss); DIN EN ISO 10545-7 (Surface Abrasion where applicable)	REACH: Regulation (EC) No. 1907/2006 (heavy metals, pigments); FDA: applicable only for food-contact ceramic decoration
Metal Decoration Ink	IS 15495; IS 6931	ISO 12647 Series; ISO 2846 Series; ISO 2409 (Adhesion),ISO 2834	ASTM D3359 (Adhesion); ASTM D523 (Gloss); ASTM D4060 (Abrasion Resistance); ASTM B117 (Corrosion Testing where metal protection is required),ASTM D1310; ASTM D1475; ASTM D2066; ASTM D4040; ASTM D4302	DIN EN ISO 2836 (Chemical Resistance of Prints); DIN ISO 12647; DIN ISO 13655; DIN EN ISO 2409 (Adhesion); DIN EN ISO 9227 (Corrosion Testing where protective coatings apply)	REACH: Regulation (EC) No. 1907/2006 (metal pigments, solvents, additives); FDA: 21 CFR 175.300 where used on food-contact metal packaging
Glass Printing Ink	IS 15495; IS 6931	ISO 2409 (Adhesion); ISO 2813 (Gloss); ISO 12647 Series,ISO 2834	ASTM D3359 (Adhesion); ASTM D523 (Gloss); ASTM D4060 (Abrasion Resistance); ASTM C1585 (Surface Absorption where applicable),ASTM D1310; ASTM D1475; ASTM D2066; ASTM D4040; ASTM D4302	DIN EN ISO 2409 (Adhesion); DIN ISO 12647; DIN ISO 13655; DIN EN ISO 2813 (Gloss); DIN EN ISO 2836 (Chemical Resistance)	REACH: Regulation (EC) No. 1907/2006 (pigments, heavy metals, additives); FDA: applicable only for food-contact glass decoration
Plastic Printing Ink	IS 15495; IS 6931	ISO 2409 (Adhesion); ISO 12647 Series; ISO 4892 Series (Weathering where applicable),ISO 2834	ASTM D3359 (Adhesion); ASTM D256 (Impact Resistance where substrate testing applies); ASTM D2244 (Colour Difference); ASTM D5264 (Abrasion Resistance),ASTM D1310; ASTM D1475; ASTM D2066; ASTM D4040; ASTM D4302	DIN EN ISO 2409 (Adhesion); DIN ISO 12647; DIN ISO 13655; DIN EN ISO 2836; DIN EN ISO 4892 (Weathering Resistance)	REACH: Regulation (EC) No. 1907/2006 (pigments, solvents, additives); FDA: applicable only for food-contact plastic printing
Conductive Ink	IS 15495; IS 6931	ISO 17009 (Conductive Materials where applicable); ISO 12647 Series where printing process applies,ISO 2834	ASTM F390 (Sheet Resistance Measurement); ASTM D257 (Electrical Resistance of Insulating Materials); ASTM B193 (Electrical Resistivity of Conductive Materials); ASTM D3359 (Adhesion),ASTM D1310; ASTM D1475; ASTM D2066; ASTM D4040; ASTM D4302	DIN EN IEC 60093 (Volume Resistivity); DIN EN IEC 62631 series (Dielectric Properties); DIN EN ISO 2409 (Adhesion); DIN ISO 12647 (Print Quality where applicable)	REACH: Regulation (EC) No. 1907/2006 (silver, copper, carbon materials, solvents); FDA: —
Functional Ink	IS 15495; IS 6931	ISO 12647 Series; Application-specific electrical/material ISO standards,ISO 2834	ASTM D2066 (Ink Testing); ASTM D3359 (Adhesion); ASTM D4060 (Abrasion Resistance); Application-specific ASTM electrical/thermal/barrier test methods,ASTM D1310; ASTM D1475; ASTM D4040; ASTM D4302	DIN ISO 13655 (Spectral Measurement); DIN ISO 12647 (Process Control); DIN EN IEC 60093 (Electrical Resistivity where conductive); DIN EN ISO 2409 (Adhesion); application-specific DIN	REACH: Regulation (EC) No. 1907/2006 (functional additives, conductive materials, pigments); FDA: application-dependent

Product / Application	BIS	ISO	ASTM	DIN	Other Standards(SSPC,AMPP, REACH, FDA)
				EN IEC standards	
Invisible Security Ink	IS 15495; IS 6931	ISO 12647 Series; ISO 2846 Series,ISO 2834	ASTM D2066 (Ink Evaluation); ASTM E308 (Colour Measurement); ASTM D2244 (Colour Difference); ASTM D3359 (Adhesion),ASTM D1310; ASTM D1475; ASTM D4040; ASTM D4302	DIN ISO 13655 (Spectral Measurement); DIN EN ISO 11664 (Colour Measurement); DIN ISO/IEC 15416 (Barcode/Verification where applicable); application-specific security printing standards	REACH: Regulation (EC) No. 1907/2006 (special pigments, fluorescent compounds, additives)
Magnetic Ink	IS 15495; IS 6931	ISO 12647 Series; ISO magnetic property testing standards where applicable,ISO 2834	ASTM A596 (Magnetic Properties of Materials where applicable); ASTM D2066 (Ink Testing); ASTM D2244 (Colour Measurement); ASTM D3359 (Adhesion),ASTM D1310; ASTM D1475; ASTM D4040; ASTM D4302	DIN EN 60404 series (Magnetic Materials); DIN EN ISO 11664 (Colour Measurement); DIN ISO 13655 (Print Measurement); DIN ISO 12647	REACH: Regulation (EC) No. 1907/2006 (iron oxide, magnetic particles, additives)

Source field: Dyes and Pigments